Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Yaponomics
Project / Solution Name	Yapoz
Name (Leader)	Abdulrahman Imaad Mohamed
Email & Phone (Leader)	swe2409099@xmu.edu.my, +60173793811
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://yapoz.vercel.app (judges: use the one-click demo buttons on the login page)
Code Repository URL (public / access granted to judges)	https://github.com/Ramilbey/yapoz

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

On WhatsApp, Telegram and Instagram, customers write to a business. Each platform pushes a signed webhook to a Next.js application deployed on Vercel which verifies it and stores the event as a Job in PostgreSQL (Supabase). The job follows in the response and any jobs that fail to run during the sweep are retried by it.

The AI agent is a tool-calling loop that is routed through Groq (OpenAI gpt-oss models) and Google Gemini, and falls back automatically. It will only respond to products on the business's menu, stocks, hours, policies and payment accounts, which are stored in the database. Orders are priced per code, are saved as pending and a receipt for payment is requested.

Photos are downloaded from channel, read by a Gemini vision model and matched against the order by a rule engine. The order is visible by the owner, the receipt and each check are displayed on the dashboard (Next.js and React) and the owner confirms or declines. The customer is informed on the same chat, stock is updated and an activity log is created. PostgreSQL LISTEN/NOTIFY is used to provide live dashboard updates. Credentials were stored in an encrypted manner. The block diagram is the same as shown in the project proposal – Figure 1.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 7.0.2	Language for the whole codebase	Apache-2.0
Next.js 16.4.2	Web app, API routes, server rendering	MIT
React and React DOM 19.3.1	User interface	MIT
Tailwind CSS 4.3.3	Styling	MIT
Radix UI (dialog, dropdown, popover, switch, tabs, tooltip) 1.x to 2.x	Accessible interface components	MIT
SWR 2.5.1	Client data fetching and live refresh	MIT
Drizzle ORM 0.45.4, Drizzle Kit 0.31	Database access and migrations	Apache-2.0, MIT

Name & Version	Purpose in Project	License
postgres (postgres.js) 3.4.9	PostgreSQL driver	Unlicense
Zod 4.6.7	Validation of every API request	MIT
jose 6.2.13, bcryptjs 3.0.3	Signed sessions, password hashing	MIT, BSD-3-Clause
grammY 1.46.0	Telegram bot framework	MIT
OpenAI SDK 7.39.0	Client for OpenAI-compatible AI APIs (Groq, Gemini)	Apache-2.0
Anthropic SDK 0.135.0	Optional Claude provider (not enabled in production)	MIT
unpdf 1.9.1, pdf-lib 1.17.1, qrcode 1.5.4	Read PDFs, generate sample PDFs, QR codes for customer links	MIT
date-fns 4.4.0	Dates and time zones	MIT
Three.js 0.186.1, React Three Fiber 9.8.2, drei, postprocessing	3D landing page	MIT (postprocessing: Zlib)
GSAP 3.15.0, Motion 14.4.0, Lenis 1.3.27	Animation and smooth scrolling	GSAP "no charge" license, MIT
Fontsource: Bricolage Grotesque, JetBrains Mono, Instrument Serif	Web fonts	OFL-1.1
tsx 4.23, ESLint 10, Playwright 1.64	Development tooling: scripts, linting, screenshots	MIT, Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Groq API (Groq)	AI answers, message classification and tool calling (OpenAI gpt-oss and Qwen models)	free
Gemini API (Google)	Reading payment receipts and menu photos (vision), chat fallback	free
WhatsApp Cloud API (Meta)	Receive customer messages, send replies, download receipt files (Meta test number)	free
Telegram Bot API (Telegram)	Receive customer messages, send replies, download receipt files	free
Instagram Graph API (Meta)	Instagram direct messages, only if the business connects it	free
Google Calendar API (Google)	Interview scheduling in the optional hiring module, off unless connected	free
Anthropic, OpenAI, OpenRouter and Ollama APIs	Supported by the AI router but disabled in production: no keys are set	not used

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel (Vercel Inc.)	Hosts the web app and API as serverless functions, runs the daily retry sweep. Deployed from the GitHub main branch	Hobby (free), functions in Washington D.C. (iad1)
Supabase (PostgreSQL)	Database, job queue, live dashboard updates	Free tier, Seoul (aws-0-ap-northeast-2)
GitHub	Source code hosting; pushes to main trigger the Vercel deployment	Free
Replit	Used during development for the initial project structure; the app is not deployed there	Sponsor-provided, not a deployment region

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-oss-120b and gpt-oss-20b	OpenAI, served by Groq	pre-trained, used as is	Apache-2.0
Qwen (fallback only; the version is chosen automatically from Groq's list)	Qwen team (Alibaba), served by Groq	pre-trained, used as is	Qwen open-weight license as published on Groq's model page
Gemini 3.x Flash and Flash-Lite (newest stable version chosen automatically, for example gemini-3.8-flash)	Google, Gemini API	pre-trained, used as is	Google Gemini API terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Forno Sette demo business: menu, prices, hours, FAQs, bookings, orders, 296 demo conversations	Created by the team (seed scripts in the repository)	synthetic	Created by the team (Apache-2.0 with the repository)	N
Kasih Hair Studio sample business	Created by the team (seed scripts in the repository)	synthetic	Created by the team	N
Sample receipts and CV PDFs	Generated by the team's scripts	synthetic	Created by the team	N
Test conversations on the live deployment	Messages sent by team members while testing	private, self-collected	Not shared	Y (display names, chat IDs, a WhatsApp number)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic), with Claude Sonnet 5.5 and Opus 5.5	Writing and testing code and database migrations, debugging, and drafting the hackathon documents	major
Replit Agent	Initial project structure	moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-written code is kept only after the TypeScript check, the production build and the automated test suite (110 tests, run on a throwaway database with the AI stubbed) pass. The team also tests every change on the live app by chatting as a customer on WhatsApp and Telegram, and reviews each change before it is committed and deployed. Facts, prices and payment details shown to customers come from the owner's data, never from model memory. The team is responsible for the code, licences and security, and can explain any part of it.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
XMUM AI Club (SDC Hackathon)	Replit (Core plan and Agent credits)	Replit Agent produced the initial project structure. The credits ran out, so development continued with Claude Code (screenshots of the depleted credits are kept as proof).

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No. Replit produced only the initial project structure. When its credits ran out the team continued in Claude Code, and the app runs on Vercel and Supabase, so nothing in the core functionality depends on a sponsor resource and it could be replaced. RunPod credits were not used because the team neither trains nor hosts a model.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Yapoz stores what customers send to a business: display name or username, chat ID or WhatsApp number, message text, photos and PDFs (including payment receipts), and the orders and bookings made from them. Message text and receipt images are sent to Groq and Google Gemini to write answers and read receipts. The team does not train models on this data.

Business credentials are encrypted (AES-256-GCM), passwords are hashed with bcrypt, every record belongs to one business workspace, and receipts open only for a signed-in owner or service user. There is no automatic deletion of chat messages yet; CVs in the optional hiring module are deleted after a number of days the owner sets. The live deployment holds only test conversations from team members; everything else is synthetic.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination: the assistant can only state facts returned by tools that read the owner's data, prices are computed by code, and unanswered questions go to a Gaps list for the owner. **Receipt reading** can misread an image or be fooled by an edited one. The AI cannot verify a payment, so its output is labelled advice, the owner confirms every order, and text printed on a receipt is treated as data, not instructions.

Provider dependency and rate limits: Groq and Gemini limit usage and change models, so the router falls back between providers and models and retries failed jobs. If all fail, the customer gets a holding message and staff are alerted. **WhatsApp:** replies are only possible within 24 hours of a customer message, and test-number access tokens expire.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: Abdulrahman Imaad Mohamed (Team Leader)

Date: 24/10/2016